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The Impact of "Black Swan" Events on Oil and the European Capital Markets

Abstract. *The price of oil and, indirectly, the performance of the stock indexes were significantly impacted by the evolution of the energy market in the context of the switch to renewable energy sources and the modifications in global energy policy. This paper proposes a comprehensive analysis of several crucial financial field elements, with particular attention to the evolution of stock indices, financial market volatility, and investor response to major events that have recently caused changes in the global economy. Using advanced economic and statistical models, the paper shows the complex relationship between the analysed stock indices, the trends and changes that occurred within them and the relevant influencing factors, such as the price of oil and the Baltic Dry Index.*

The results of the DCC-GARCH models highlight a significant positive correlation between the stock indices of several European countries, underlining the interconnection of these markets in conditions of increased volatility.

Keywords: *correlation, impact of events, oil price fluctuations, DCC-GARCH model, stock indexes, energy market.*

JEL Classification: C01, C12, C32, C38, C53, C55, C58, C61, C81.

1. Introduction

In the current context marked by changes and economic uncertainty, oil price fluctuations are of great importance, both in that they can significantly influence a series of economic indicators whose instability can cause significant financial losses in the economy, as well as for investors. With the rapid development of international relations, in a constantly changing economic context, the interconnection and influences between economic and financial markets highlight the particular

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importance for investors in search of risk hedging strategies and portfolio diversification. With all the rapid changes in the world of oil and its replacement by renewables, oil is an important part of the economy, a necessary commodity for global economies, as well as an important stock market investment tool, and investors want to closely monitor oil prices and the changes that replace it with renewable resources to make informed decisions about their portfolios.

Contemporary events such as medical, geopolitical, or governmental conflicts that have reduced production in the extraction areas have produced strong changes in the price of oil and fluctuations in the prices of stock indices.

The purpose of the paper is to observe the impact of the negative events that have affected the global economy in recent years, to analyse of oil price variations, considering their importance, and to investigate the influence these fluctuations have on the financial markets. The events of recent years make us notice that major changes are observed on the oil market during the COVID-19 pandemic because of the closure of industry and the population during the (lockdown) period, but also of the decrease in mobility, and as such there is a decrease in demand for oil. What made this market unbalanced was the fact that oil production did not decrease at the beginning of the COVID-19 pandemic; on the contrary, it remained the same until the (lockdown) period, a fact that led to an oversupply on the market when many of the oil depots storage became full, and as a result in April 2020 oil prices drop a lot.

The restoration of the oil market and the management of market volatility was done quite quickly with the help of the Organization of Petroleum Exporting Countries (OPEC), which intervened to adjust production. The major changes in the oil market during the COVID-19 pandemic, directly affected the capital market and economic sectors related to the oil industry, and this was best observed when oil prices fell significantly due to the sudden reduction in demand and the stock markets have been negatively affected. The increased volatility in the stock markets because of the uncertainty existing during the COVID-19 pandemic until the discovery of the vaccine, as well as the economic stimulus measures, produced important fluctuations in the prices of the stock indices. Thus, it is found that the increased volatility of oil prices in recent years is due to numerous factors that produced it, including instability in important regions with concentrated oil resources, the COVID-19 pandemic that created production constraints, energy changes, or armed military conflicts. The main objective of the paper is to analyse the recent disturbing episodes, which have affected the price of oil and the return of the stock indices proposed for analysis. The war between Russia and Ukraine affected the oil industry and the energy market in Europe and contributed to the volatility of oil and natural gas, causing price increases on international markets. Russia, one of the world's largest producers and suppliers of oil and natural gas, because of the sanctions and restrictions imposed due to the invasion of Ukraine, has reduced its oil and gas transport to many European countries to the point of cessation. Globally, as Russia has an important influence on energy markets, a disruption of this kind has

significant consequences and can affect national economies, creating pressure on financial stability. At the same time, the armed conflict between Russia and Ukraine has negative consequences on the capital markets as well, and here we can observe the volatility of the financial market, and the damage to investors' confidence due to the geopolitical conflict, led them to look for assets considered safe, such as gold or government bonds, behaviour that influences the direction of stock markets. Events of this kind are called by the analyst Taleb (2008) "black swan" and refer to unpredictable, rare events with significant consequences and massive impact and are considered inevitable and difficult to anticipate. As such, the paper traces the "black swan" periods that had a stronger impact, observing the volatility of oil prices and stock indices during the crisis period of 2008 and especially in the current period when the global economy faced a succession of such events that appeared out of the blue but that changed the economic and financial landscape.

The paper manages to highlight oil volatility and investigate how it influences financial markets using well-known econometric models such as multivariate series analysis and the DCC-GARCH model. The importance of the study, in the current context, consists in highlighting the volatility of oil and stock indices, volatility with a significant impact on the main economic indicators and whose instability can lead to large financial losses for the economy and investors. Therefore, this paper is important for investors in making well-informed investment decisions and anticipating risks. The novelty of the work consists in the interconnection between Brent Oil prices, the Baltic Dry index (BDI) and various European stock market indices such as those on: the Romanian stock market (BET index), the Polish stock market (WIG index), the French stock market (CAC40 index), the German stock market (DAX index), the Austrian stock market (ATX index), the Dutch stock market (AEX index) and the English stock market (FTSE100 index) where it is highlighted how the volatility of the oil market can affect the financial market, when there are impacts of negative events such as "black swan".

The analysis covers approximately 15 years, from January 2008 to November 2023, and the results obtained are fully consistent with previous research and confirm the presence of volatility in these markets. Analysing the intricate relationships between Brent Oil prices, the Baltic Dry Index (BDI) and selected European stock indices, this paper provides a detailed insight into how oil market volatility can affect the performance of financial markets. The results obtained provide investors with a valuable mechanism to better understand the risk associated with the oil market and to identify investment opportunities in a volatile environment and provide important steps in making more informed and accurate investment decisions. By looking at current events and their impact on financial markets, the study provides a comprehensive analysis of trends and influences that can lead to more informed decisions in the future.

To achieve the set objectives, this paper is divided as follows: literature review, methodology, results and discussion, and conclusions.

2. Literature review

The specialised literature highlights the factors and volatility of oil, as well as its impact on the performance of the financial market. Some significant works on the volatility of oil prices in the field are the work of Dvir and Rogoff (2009), which shows both the dynamic behaviour of oil prices and the fact that shocks on the oil market have different effects on the real price of oil not because of supply or demand, but because of the ability (or lack thereof) of key market players to restrict access to supplies. Therefore, effective restrictions on access to excess supply can generate oil price growth, shocks that are persistent, and with increased volatility. In their remarkable study, Fedorova and Saleem (2010) shed light on the complexity of volatility in Central Eastern European markets. In a particularly detailed and meticulous effort, the authors analyse four key European states, and by exploring unidirectional and bidirectional linkages, the study reveals the spread of volatility in these countries. With a multidirectional approach, the researchers investigate the mutual influence between financial markets and currency markets in each country in part, and to examine the transmission of shocks and volatility, the authors use an unrestricted bivariate GARCH-BEKK (1,1) model, thus providing a deep insight into how changes in one domain can affect and shape behaviour in another domain. Through this detailed and rigorous research, Fedorova and Saleem (2010) make significant contributions to understanding the complex interactions between the financial and foreign exchange markets of these countries, thereby providing a solid basis for further analysis and decision-making in the context of market volatility.

Contagion and volatility between crude oil and financial markets is investigated by Chatziantoniou et al. (2021) through a standard vector autoregressive with time-varying parameters (TVP-VAR), and the results of their study suggest that the crude oil market is an important transmission network for volatility shocks.

In their work Abhyankar, Xu and Wang (2013) using a structural vector autoregressive model (SVAR), show the relationship between oil price shocks and the Japanese stock market, and according to the results obtained, it is found that the oil price shocks arising from changes in global aggregate demand are positively correlated with the return of the Japanese stock market, which means that an increase in oil prices is not always bad news for the Japanese stock market, as it reacts negatively to increases in oil prices related to specific oil market demand shocks.

In his study, Abdollah (2023) examines oil price volatility by forecasting it, using information from news and Twitter feeds. This study highlights how oil price fluctuations are heavily influenced by factors such as geopolitical conflicts and natural disasters, which are often widely publicised and extensively discussed in the media. Therefore, this research reveals the complex links between significant real-world events and oil market volatility, making a significant contribution to understanding this phenomenon.

The paper written by Lovcha and Perez-Laborda (2022) investigated oil volatility using a FIVAR model. Research results indicate that oil volatility does not exhibit persistence and is characterised by a combination of short and long memory.

The long-memory component gives the volatility series significant persistence. Related measures have shown that while oil volatilities are significantly correlated, they are much less interconnected than standard VAR models suggest, especially at low frequencies. This research provides a deep insight into the behaviour of oil volatility, making significant contributions to understanding the complexity of this phenomenon in the context of financial markets. Research using short-memory multivariate GARCH models by Marchese, Kyriakou, Tamvakis and Di Iorio (2022) focuses on the fundamental relationship between crude oil prices and refined products. The paper in which they were compared for spot yields, three major oil markets examine the potential benefits of using fractionally integrated multivariate GARCH models from a forecasting and risk management perspective. Results indicate that multivariate models incorporating long memory outperform short-memory benchmarks in forecasting the conditional covariance matrix and associated hazard magnitude. The paper makes an innovative contribution to the analysis of the relationship between crude oil and its refined products, providing refiners, physical oil traders, non-oil traders, and other agents in the energy markets with meaningful information for hedging operations and risk management.

Researchers such as Gong et al. (2020) focused on investigating the asymmetric and non-linear laws between shipping costs and financial markets, studying contagion risk in the context of extreme events, and making a significant contribution by using the switch of a Markov copula (MS) model with three variants. Their study revealed contemporaneous two-way lead-lag relationships between the freight market, as measured by the Baltic Dry Index, and the US and Chinese stock markets. The authors identified that the risk of contagion increases as the volume of shipping between the US and China. These findings highlight the importance of the subtleties and interconnections between the economic and financial sectors, bringing to the fore complicated relationships between the shipping market and stock markets.

In their work, Baumeister and Kilian (2016) bring to the fore the complexity and unpredictability of the oil market and the significant impact it has on economies and financial decisions. It is demonstrated that the magnitude and timing of price shocks depend largely on expectations regarding the evolution of oil prices. Price shocks reside in the fact that they influence the economic decisions of individuals and companies. Higher gasoline prices generate an increase in the price of oil, which reduces the income available for other activities, thus affecting household spending. Another significant impact of oil price shocks is related to expectations regarding the future of oil prices. These expectations are essential in calculating the net present value (NPV) for future investment projects that depend on the price of oil, such as car manufacturers making key decisions about new car models depending on the price of oil, and households deciding what type of car to buy, both of which are influenced by the evolution of oil prices in the future. It is observed that oil price shocks are not only limited to the current value of the price, but rather to how they change the expected course of the price in the future. These complex dynamics underscore the importance of careful analysis of the oil market and how it can affect

economic decisions and investments, thereby providing significant insight for economic and financial actors. Alamgir and Amin (2021) in their study use a Nonlinear Autoregressive Distributed Lag (NARDL) model to investigate the complex relationship between oil prices and the stock market. The results obtained reveal a strong positive relationship between the world oil price and the stock market indices, and, what is more remarkable, the response of the stock market index to positive and negative oil price shocks is asymmetric. The study shows that higher oil prices in the world market have a significant stimulatory effect on the share price. This finding not only highlights the importance of oil prices in the dynamics of financial markets, but also highlights the sensitivity to events and changes in the oil industry. Therefore, this research provides a significant contribution to understanding the subtleties of the relationship between oil prices and the behaviour of stock markets, providing a clear and detailed insight into the complex impact they have on the economy.

Prabheesh, Padhan and Garg (2020) investigated the relationship between oil price returns and stock price returns using the DCC-GARCH model for four major oil importing countries in Asia. Although the authors find that the correlations between the two price returns are small, the values are positive. Therefore, this positive move may reduce opportunities for portfolio diversification as the uncertainty associated with COVID-19 may affect economic performance considering lower oil prices, which act as a signal of future demand contraction and the associated weak economic outlook.

In their study García and Rambaud (2023) reveal an important aspect of international oil markets. These researchers emphasise that the volatility of oil prices is a complex phenomenon, generated by shocks and frictions of demand and supply, a phenomenon that amplifies in periods of international crisis. The methodology of the paper combines the analysis of volatility using the GARCH model with the impulse response method for shocks and with the index approach of spillover by Diebold and Yilmaz (2012), and the results obtained show significant increases in volatility in the months correlated with economic blockages in the US economy and with the identification of the channel of uncertainty as the main factor leading to the spread of volatility, the COVID- 19 pandemic, when price volatility in international oil markets reached unprecedented levels, surpassing even the levels recorded during the financial crisis of 2007-2009. The research makes a significant contribution to understanding the complexity of volatility in oil markets, and clearly shows how factors such as uncertainty and international crises can have a profound and lasting impact on world economies.